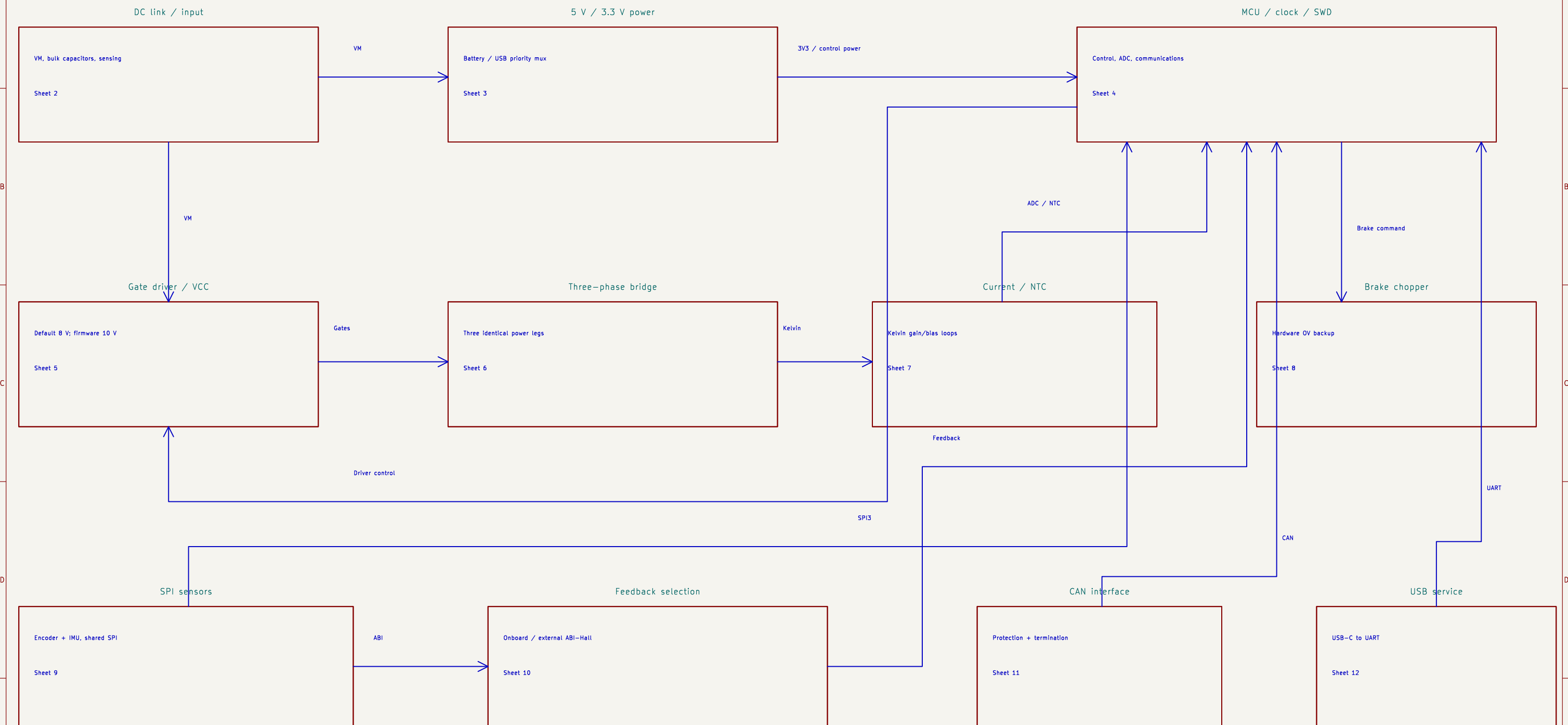


ShiroFOC – CONNECTED FUNCTIONAL SCHEMATIC

A0 electrical circuit preserved | 18-42 V target | Prototype current / thermal limits remain applicable



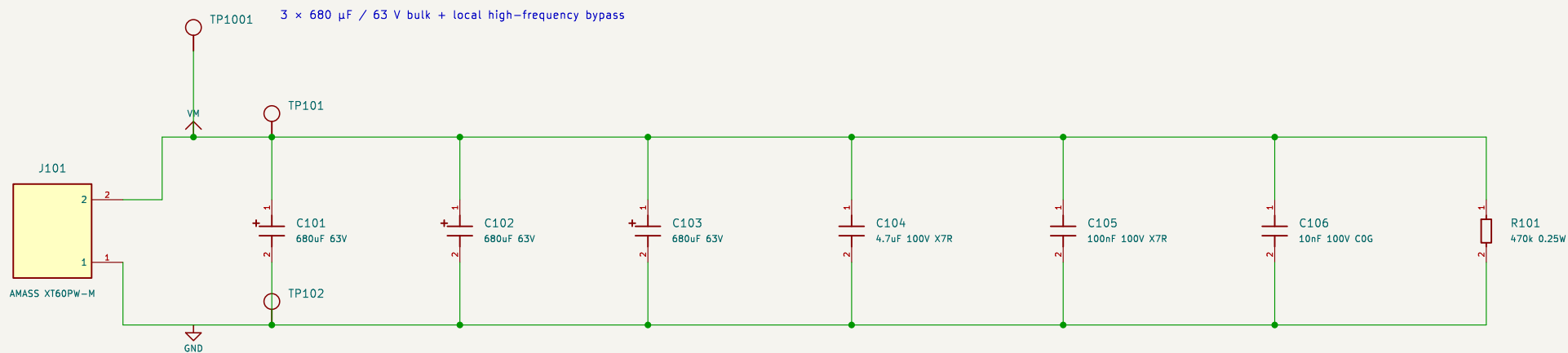
Circuit sheets use wires for local connections and named ports only at functional boundaries.
Power/ground symbols are shared rails. Dashed/graphic overview links explain functional relationships, not PCB routing.

Start with PCB_LAYOUT_HANDOFF.md and FIRMWARE_BRINGUP_CONTRACT.md.
Readability redraw does not add hardware qualification or change the power/current envelope.

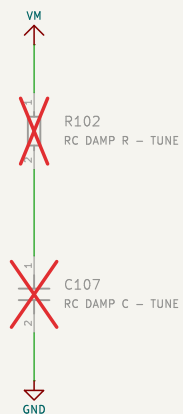
Sheet: / File: ShiroFOC_KiCad.kicad_sch		
Title: ShiroFOC A1 – connected functional overview		
Size: A3	Date: 2026-09-15	Rev: A1-Drawing
KiCad E.D.A. 9.0.4		Id: 1/12

BATTERY / DC LINK

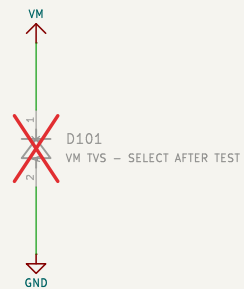
18-42 V operating bus • keyed XT60 • external fuse / precharge required



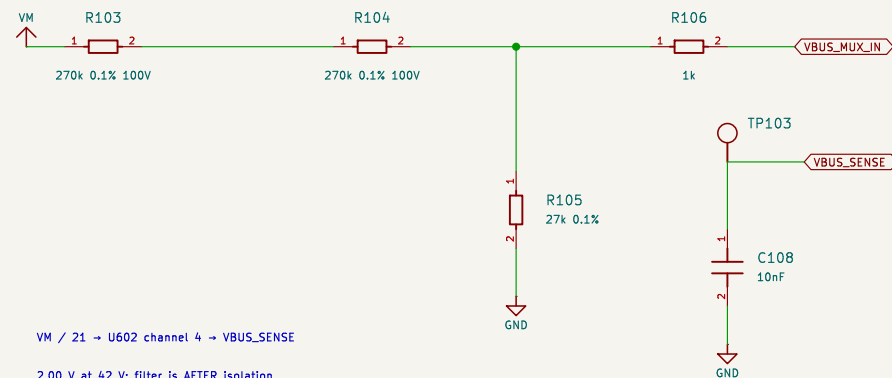
OPTIONAL INPUT DAMPING



DNP: choose clamp and damping only after cable / hot-plug tests.



BUS DIVIDER / ISOLATED ADC FILTER



VM / 21 → U602 channel 4 → VBUS_SENSE

2.00 V at 42 V; filter is AFTER isolation.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /DC link / input/

File: 01_Battery_DC_Link.kicad_sch

Title: Battery input, DC link and bus measurement

Size: A4 Date: 2026-09-15

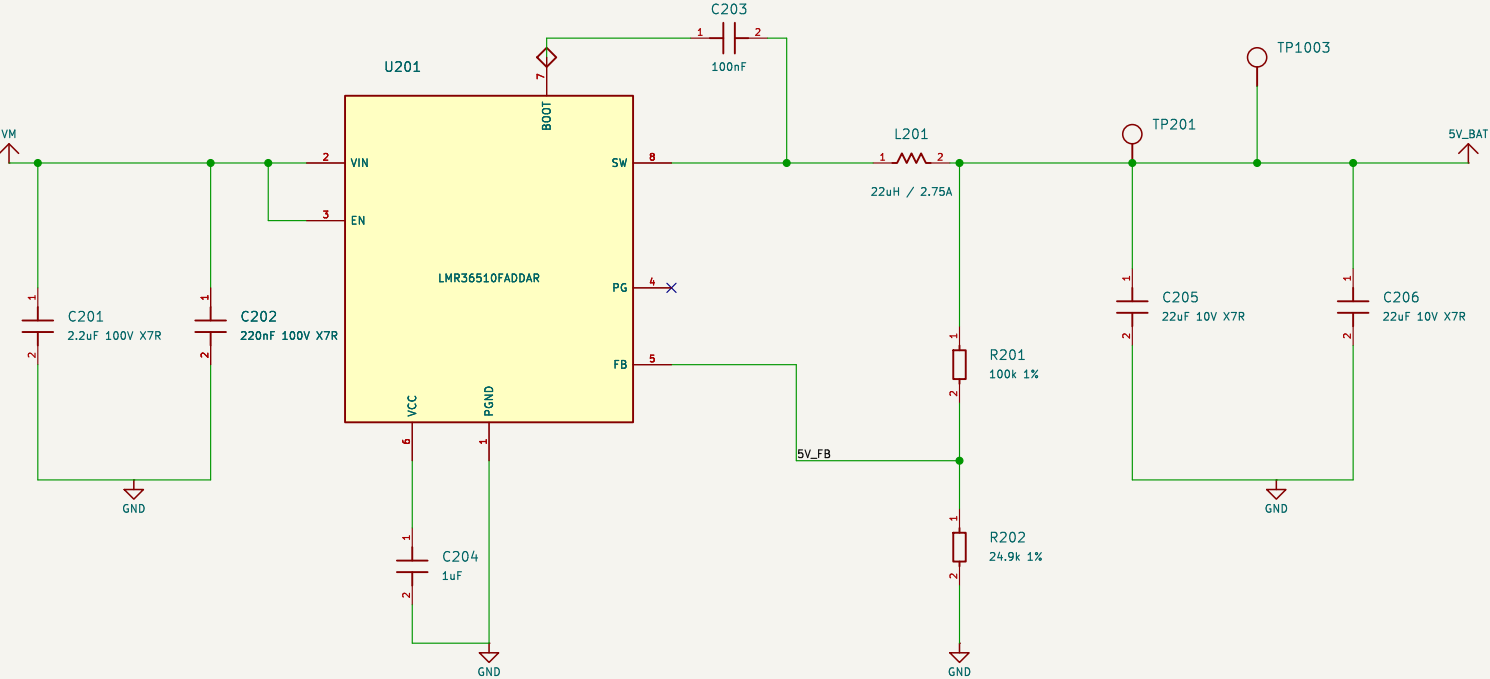
KiCad E.D.A. 9.0.4

Rev: A1-Drawing

Id: 2/12

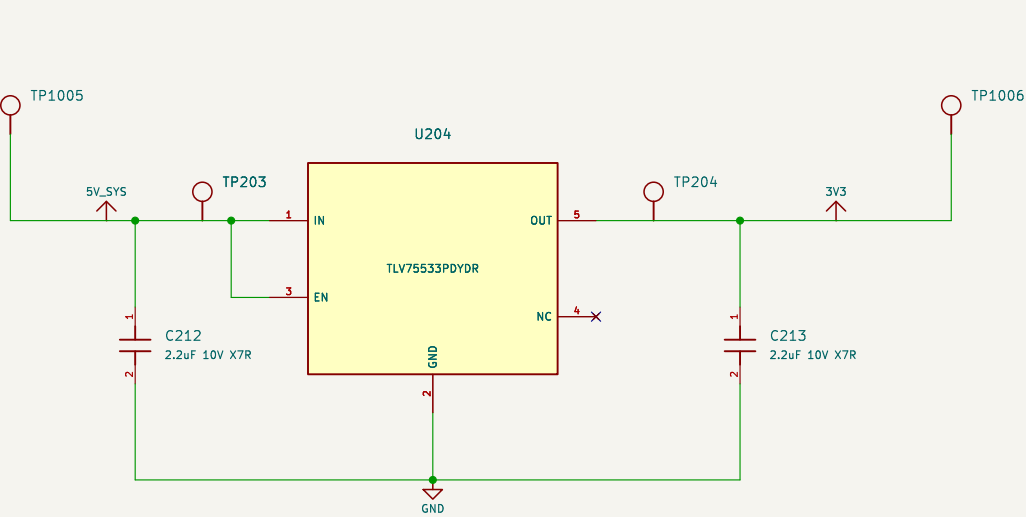
AUXILIARY POWER – USB HAS PRIORITY

BATTERY → 5 V BUCK



FB loop: 100 kΩ / 24.9 kΩ → approximately 5.02 V

5 V SYSTEM → 3.3 V

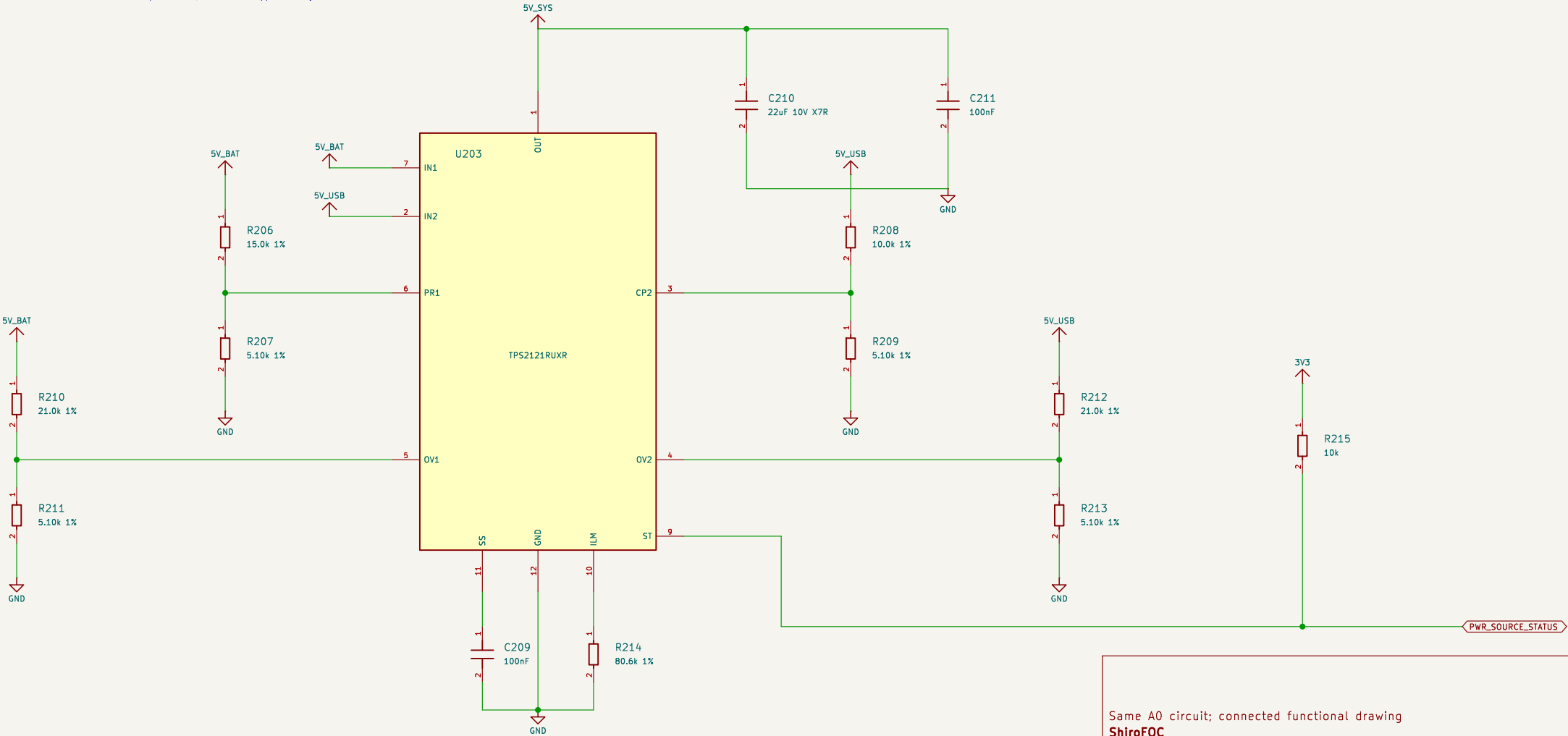


Debugger VTREF is sense-only; never inject 3V3.

U204 thermal pad / pin 2 returns to GND.

USB-PRIORITY POWER MUX

5V_USB arrives from the USB service sheet.



At 5 V: PR1 = 1.269 V; CP2 = 1.689 V.

USB wins the priority comparison. OV ≈ 5.43 V.

80.6 kΩ sets approximately 1.5 A current limit.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /5 V / 3.3 V power/

File: 02_Auxiliary_Power.kicad_sch

Title: Auxiliary supplies / USB priority

Size: A3

Date: 2026-09-15

Rev: A1-Drawing

KiCad E.D.A. 9.0.4

Id: 3/12

CONTROLLER / CLOCK / DEBUG

A

B

C

D

MCU / ANALOG SUPPLIES

E

F

A

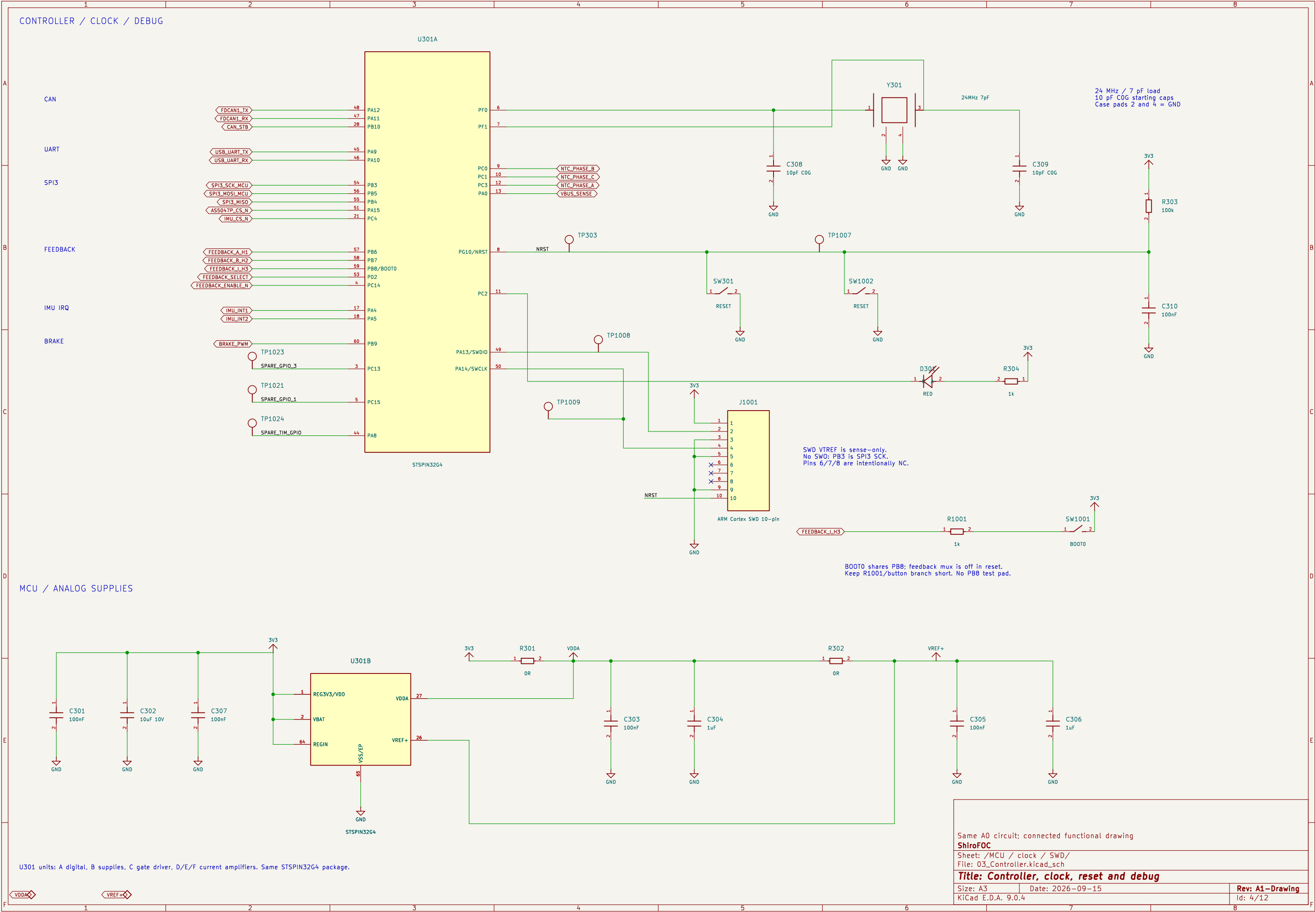
B

C

D

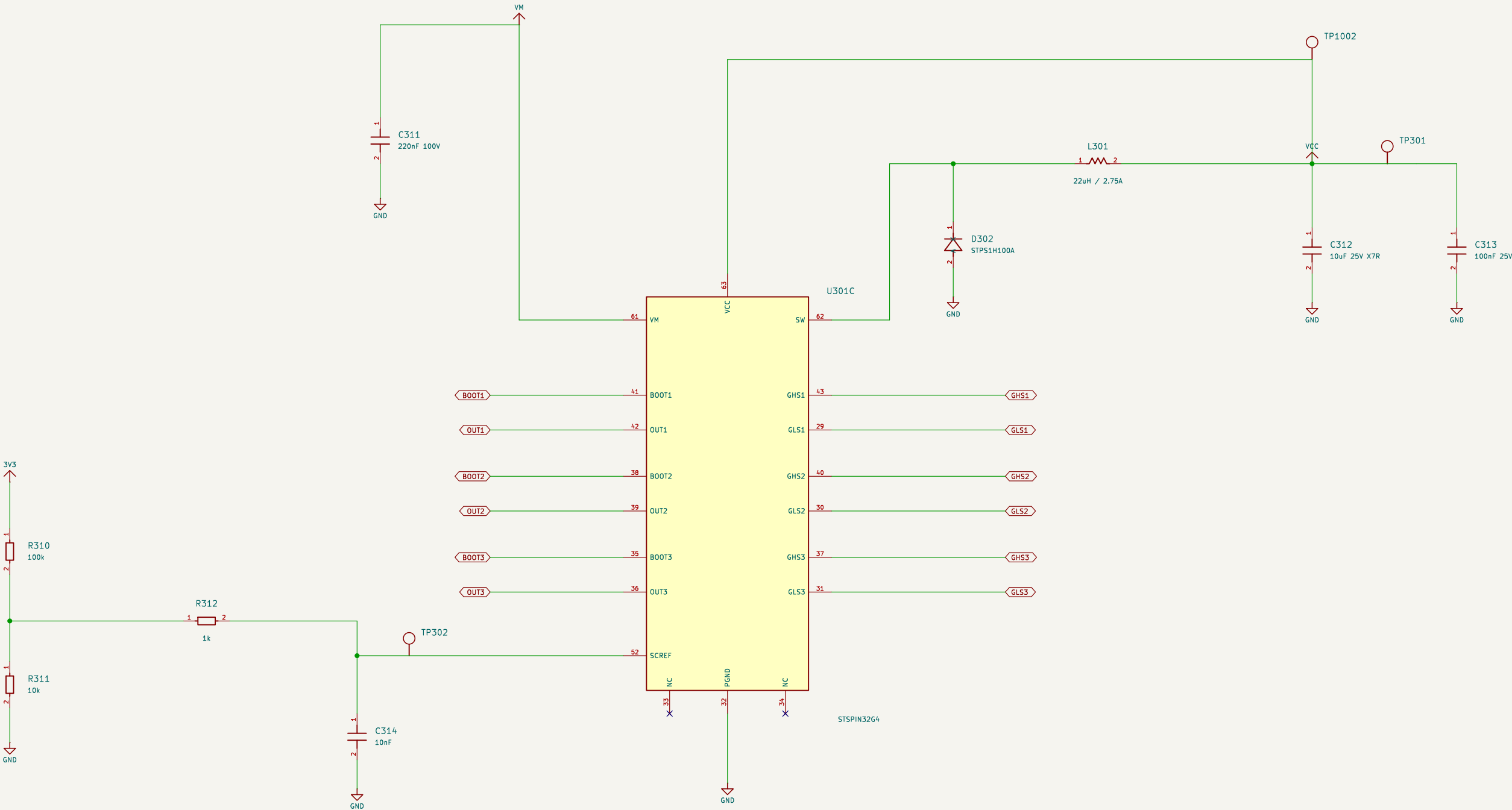
E

F



GATE DRIVER / VCC CONVERTER

VDS FAULT REFERENCE



SCREF is a VDS fault threshold (~0.30 V), not a precision phase-current limit. Configure COMP1/2/4 and TIM1 break before enabling PWM.

VCC starts at 8 V. Reapply the 10 V setting after each VM return. REGIN and REG3V3/VDD use external 3V3 (unit B). Gate supply stays off with USB alone.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /Gate driver / VCC/

File: 04_Gate_Driver.kicad_sch

Title: STSPIN gate drive and 10 V converter

Size: A3

Date: 2026-09-15

Rev: A1-Drawing

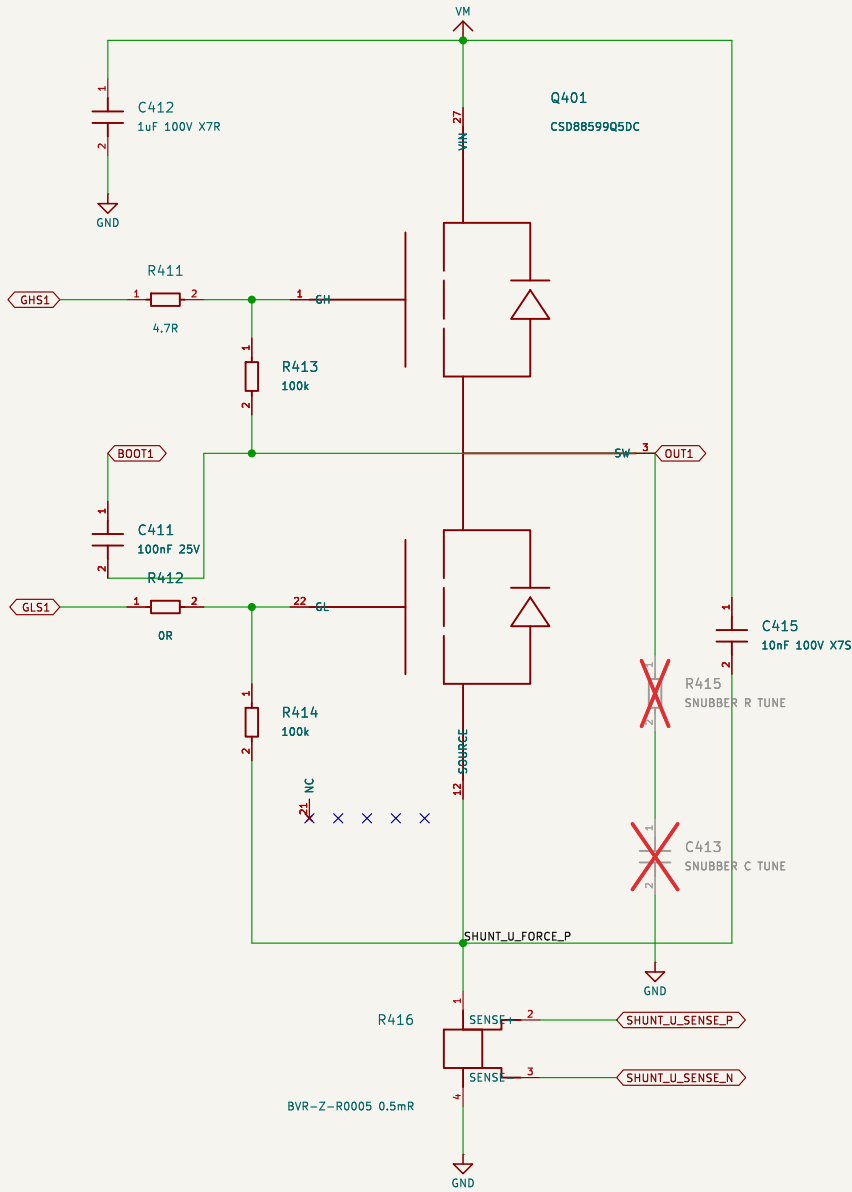
KiCad E.D.A. 9.0.4

Id: 5/12

THREE IDENTICAL PHASE POWER LOOPS

Gate resistors at MOSFET pins. The 10 nF capacitor returns above the shunt; the 1 uF capacitor returns to GND.

PHASE U

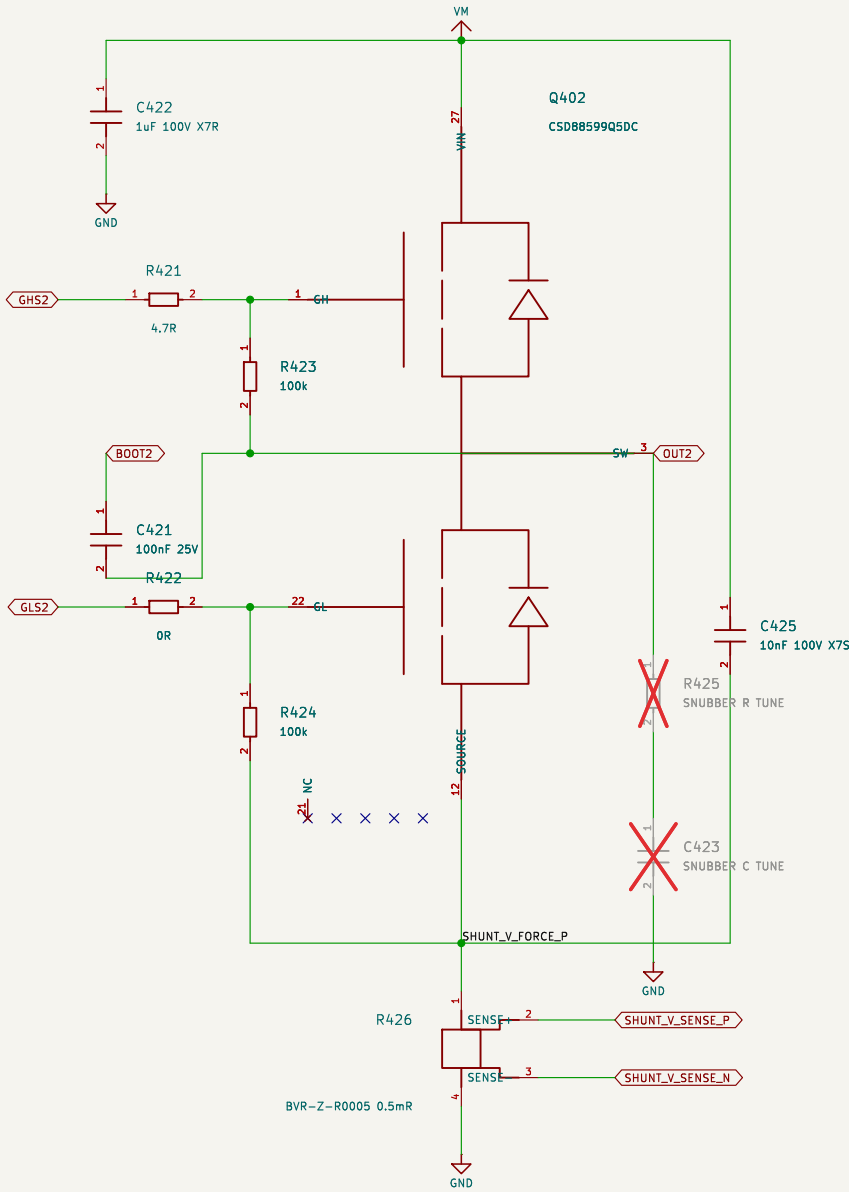


Package pins: VIN 27; GH 1; GL 22.
SW/SH: 2, 3-11; source: 12-20.
NC: 21, 23-26. Stacked pins are retained.

Kelvin pads 2/3 go only to the amplifier.
Force-current copper uses pads 1/4.
Snubber R/C are DNP tuning positions.

Layout order: VM ceramic -> high-side FET -> low-side FET -> local source return,
Keep the bootstrap and gate loops short; do not route motor current in Kelvin sense traces.

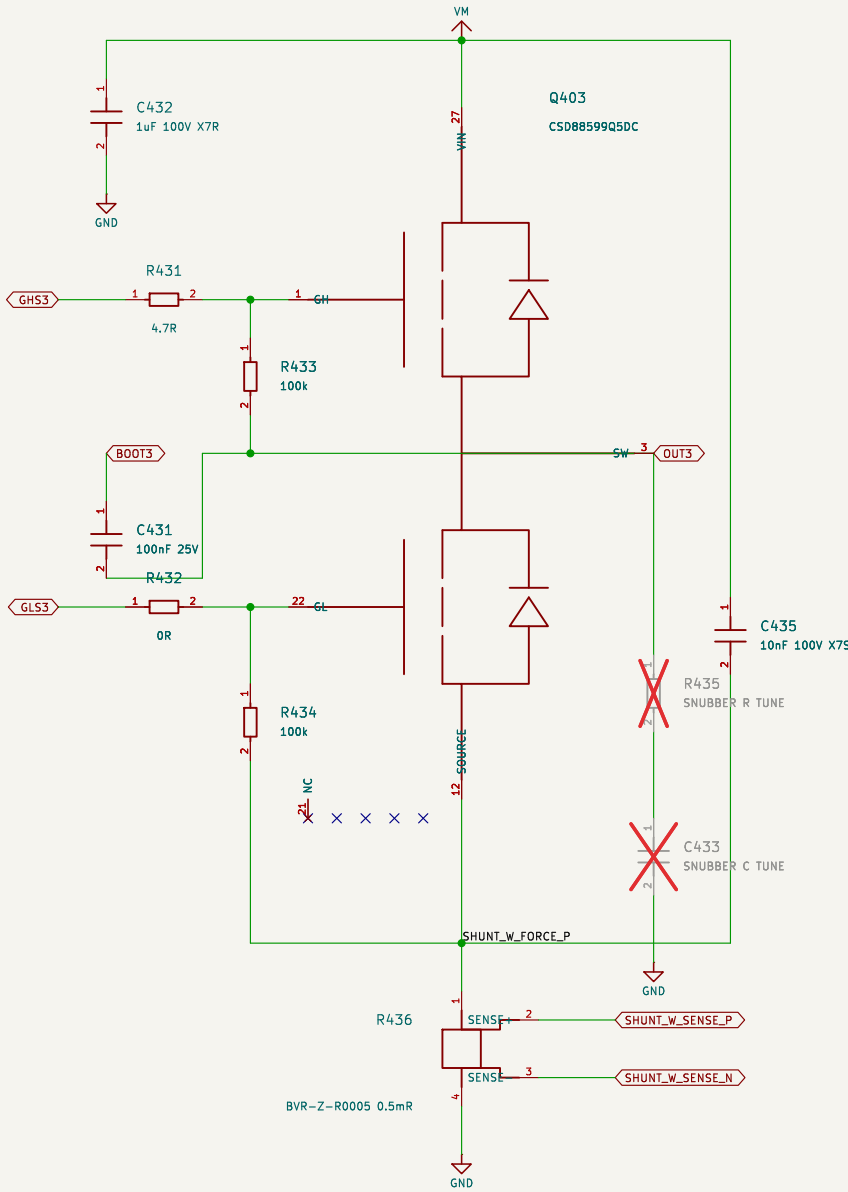
PHASE V



Package pins: VIN 27; GH 1; GL 22.
SW/SH: 2, 3-11; source: 12-20.
NC: 21, 23-26. Stacked pins are retained.

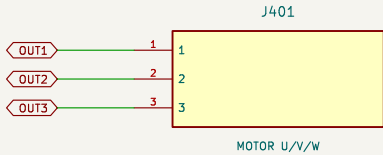
Kelvin pads 2/3 go only to the amplifier.
Force-current copper uses pads 1/4.
Snubber R/C are DNP tuning positions.

PHASE W



Package pins: VIN 27; GH 1; GL 22.
SW/SH: 2, 3-11; source: 12-20.
NC: 21, 23-26. Stacked pins are retained.

Kelvin pads 2/3 go only to the amplifier.
Force-current copper uses pads 1/4.
Snubber R/C are DNP tuning positions.



Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /Three-phase bridge/

File: 04_Inverter.kicad_sch

Title: Three-phase inverter / gate loops / Kelvin shunts

Size: A3

Date: 2026-09-15

Rev: A1-Drawing

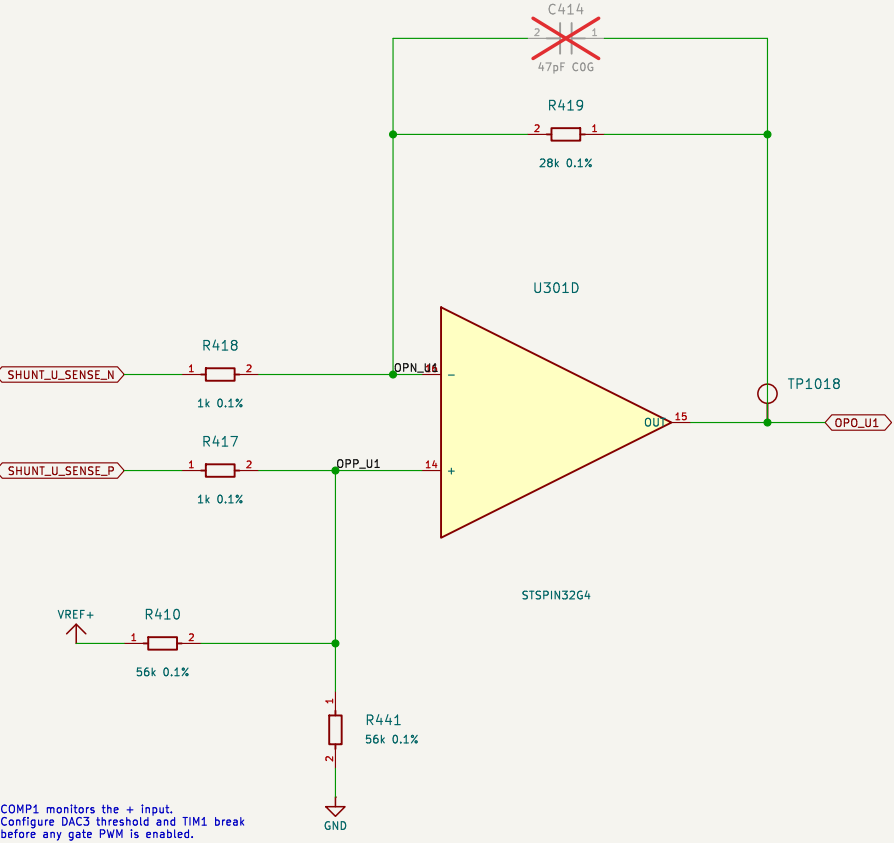
KiCad E.D.A. 9.0.4

Id: 6/12

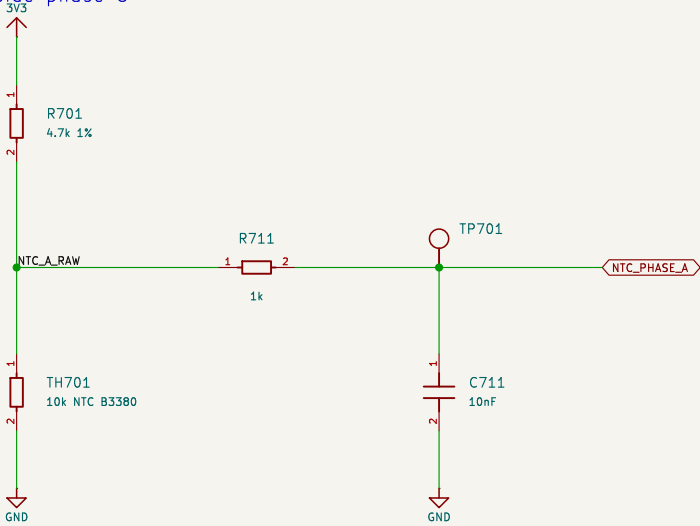
MATCHED CURRENT CHANNELS AND LOCAL MOSFET TEMPERATURE

$V_{OUT} = V_{REF}/2 + 28 \times (SENSE+ - SENSE-)$. At 0.5 mOhm: 14 mV/A. Bias resistors establish the bipolar zero-current point.

PHASE U / OPAMP1

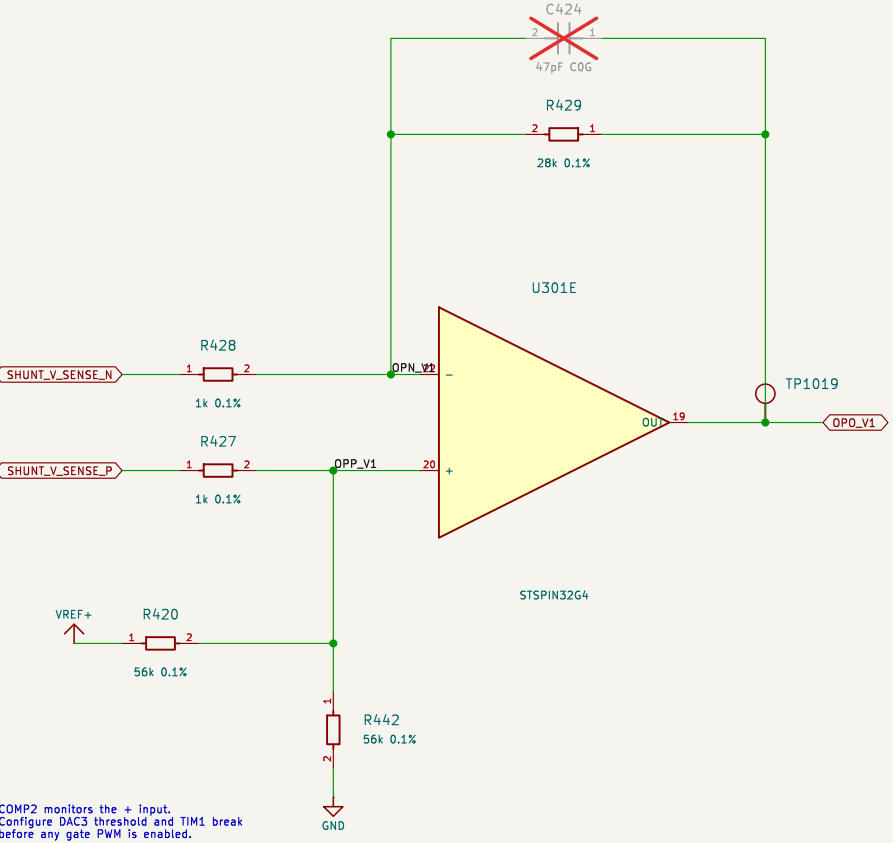


NTC A – place beside phase U

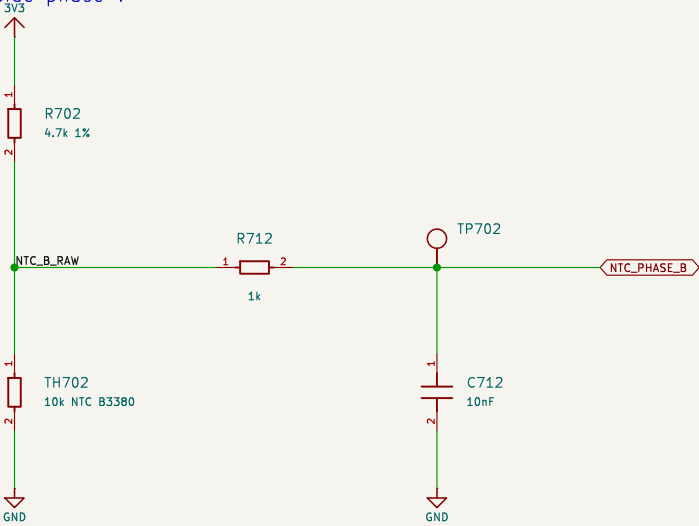


NTC faults: open -> ADC near 3V3; short -> ADC near GND. Use the hottest valid channel and calibrate protection against measured temperature.

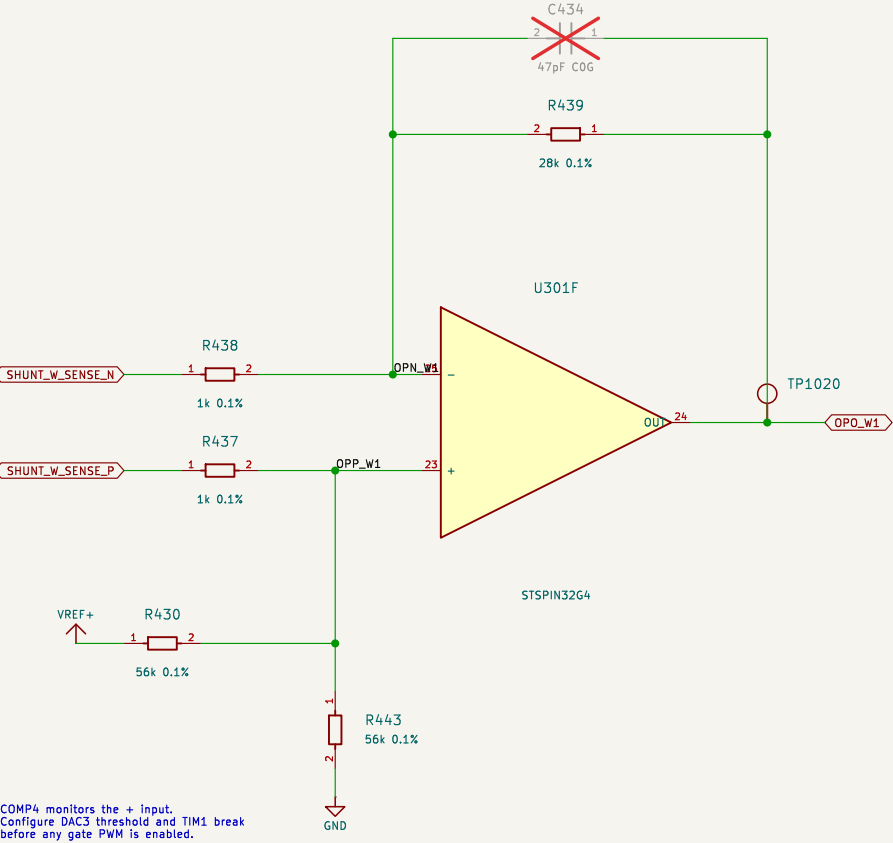
PHASE V / OPAMP2



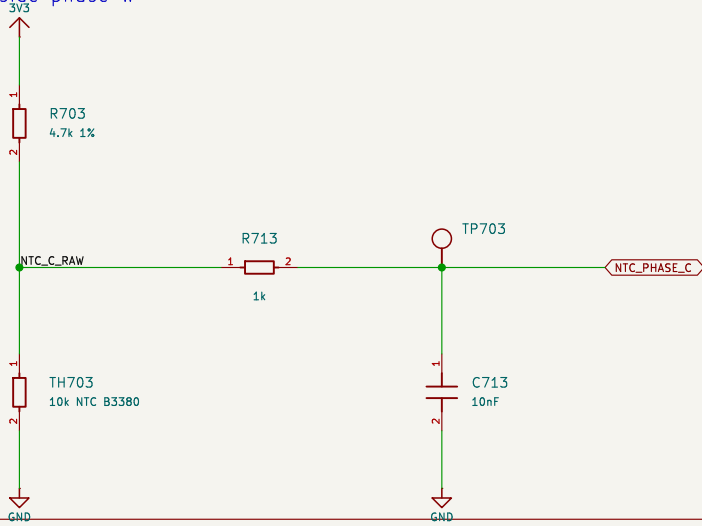
NTC B – place beside phase V



PHASE W / OPAMP3



NTC C – place beside phase W



Same A0 circuit; connected functional drawing		
ShiroFOC		
Sheet: /Current / NTC/ File: 05_Current_Temperature.kicad_sch		
Title: Three Kelvin current amplifiers / phase temperature		
Size: A3	Date: 2026-09-15	Rev: A1-Drawing
KiCad E.D.A. 9.0.4		Id: 7/12

BRAKE CHOPPER – TWO COMMAND SOURCES, ONE GATE DRIVE

FIRMWARE / HARDWARE DIODE OR

INDEPENDENT OVERVOLTAGE COMMAND

LOCAL GATE DRIVE / POWER SWITCH

U502 commands braking at approximately 45.9 V rising / 44.2 V falling; MCU PWM can also command the diode OR.

Fit and rate the external resistor before regeneration. Both resistor wires carry bus voltage; pulse energy is application-specific.

Place U501 decoupling / R503 / D501 at Q501. Return the driver ground to Q501 source with a short Kelvin connection.

Same A0 circuit; connected functional drawing
ShiroFOC
 Sheet: /Brake chopper/
 File: 05_Brake_Chopper.kicad_sch
Title: Brake chopper / overvoltage protection
 Size: A3 Date: 2026-09-15 Rev: A1-Drawing
 KiCad E.D.A. 9.0.4 Id: 8/12

BRAKE CHOPPER – TWO COMMAND SOURCES, ONE GATE DRIVE

FIRMWARE / HARDWARE DIODE OR

INDEPENDENT OVERVOLTAGE COMMAND

LOCAL GATE DRIVE / POWER SWITCH

U502 commands braking at approximately 45.9 V rising / 44.2 V falling; MCU PWM can also command the diode OR.

Fit and rate the external resistor before regeneration. Both resistor wires carry bus voltage; pulse energy is application-specific.

Place U501 decoupling / R503 / D501 at Q501. Return the driver ground to Q501 source with a short Kelvin connection.

Same A0 circuit; connected functional drawing
ShiroFOC
 Sheet: /Brake chopper/
 File: 05_Brake_Chopper.kicad_sch
Title: Brake chopper / overvoltage protection
 Size: A3 Date: 2026-09-15 Rev: A1-Drawing
 KiCad E.D.A. 9.0.4 Id: 8/12

BRAKE CHOPPER – TWO COMMAND SOURCES, ONE GATE DRIVE

FIRMWARE / HARDWARE DIODE OR

INDEPENDENT OVERVOLTAGE COMMAND

LOCAL GATE DRIVE / POWER SWITCH

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U501
UCC27517ADBVR

U502
TLV3012BIDBVR

Q501
CSD19531Q5A 100V

D501
BZT52C12 12V

D502
BAT54H

D503
BAT54H

D504
B5100C 100V 5A

R501
10k

R502
33R

R503
4.7R

R504
10k

R505
10k

R506
150k 0.1% 100V

R507
150k 0.1% 100V

R508
150k 0.1% 100V

R509
12.7k 0.1%

C501
1uF 25V

C502
100nF 25V

C503
100nF

C504
1nF COG

TP501

TP502

EXTERNAL DUMP R

VM

VCC

GND

BRAKE_PWM

BRAKE_PWM_DRV

BRAKE_OV_CMD

IN+

IN-

OUT

REF

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BRAKE CHOPPER – TWO COMMAND SOURCES, ONE GATE DRIVE

FIRMWARE / HARDWARE DIODE OR

INDEPENDENT OVERVOLTAGE COMMAND

LOCAL GATE DRIVE / POWER SWITCH

U502 commands braking at approximately 45.9 V rising / 44.2 V falling; MCU PWM can also command the diode OR.

Fit and rate the external resistor before regeneration. Both resistor wires carry bus voltage; pulse energy is application-specific.

Place U501 decoupling / R503 / D501 at Q501. Return the driver ground to Q501 source with a short Kelvin connection.

U501
UCC27517ADBVR

U502
TLV3012BIDBVR

Q501
CSD19531Q5A 100V

D501
BZT52C12 12V

D502
BAT54H

D503
BAT54H

D504
B5100C 100V 5A

R501
10k

R502
33R

R503
4.7R

R504
10k

R505
10k

R506
150k 0.1% 100V

R507
150k 0.1% 100V

R508
150k 0.1% 100V

R509
12.7k 0.1%

C501
1uF 25V

C502
100nF 25V

C503
100nF

C504
1nF COG

TP501

TP502

EXTERNAL DUMP R

VM

VCC

GND

BRAKE_PWM

BRAKE_PWM_DRV

BRAKE_OV_CMD

IN+

IN-

OUT

REF

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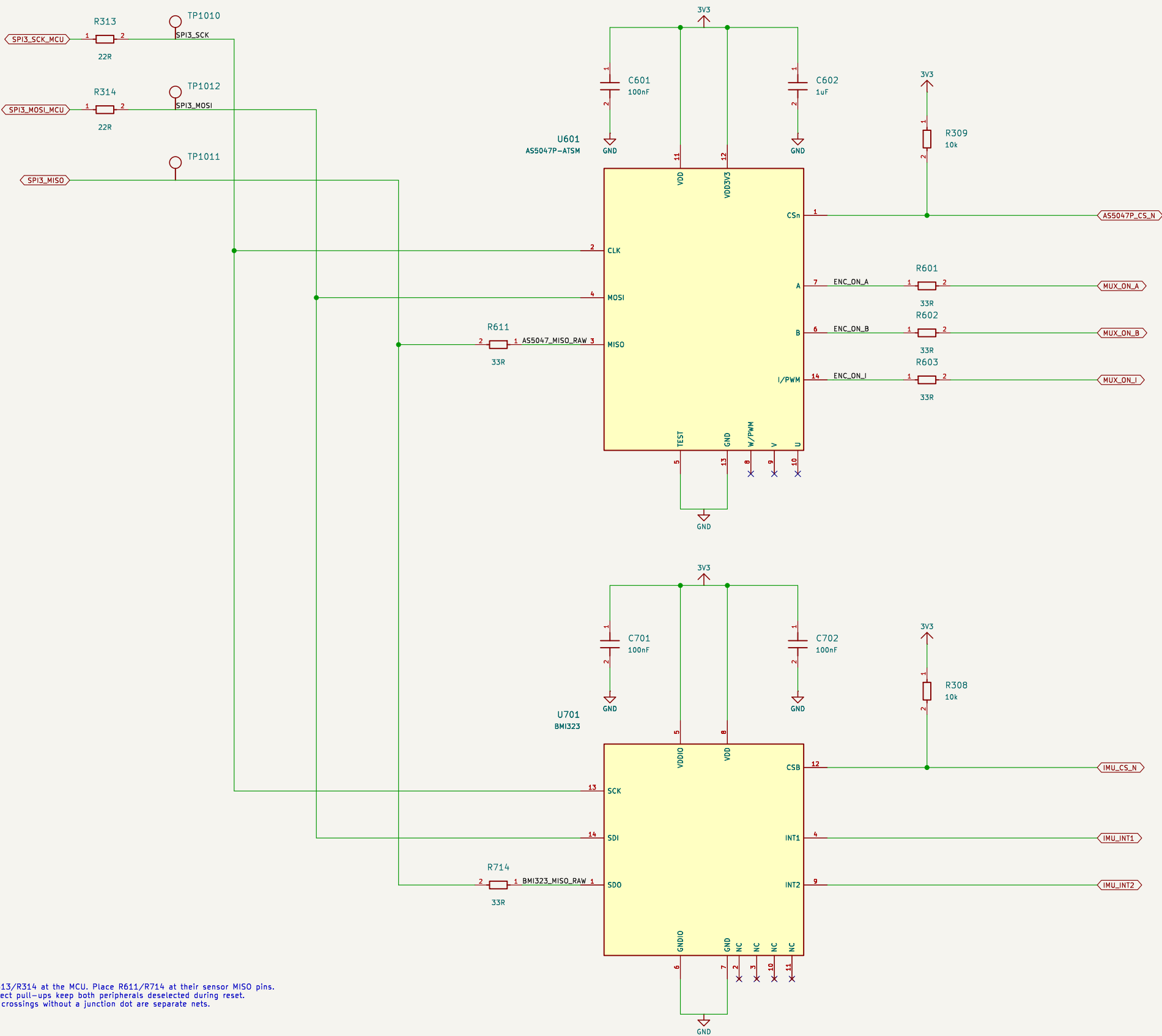
243

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Id: 8/12

SPI3: one shared bus, separate chip selects, local return damping



ONBOARD POSITION
ABI outputs feed the selector sheet.
TEST is tied low for normal operation.
Align the sensor with the shaft magnet.

IMU
Mark X/Y axes on the PCB.
Keep power-stage heat and switching
loops away from the sensor.

Place R313/R314 at the MCU. Place R611/R714 at their sensor MISO pins.
Chip-select pull-ups keep both peripherals deselected during reset.
SPI wire crossings without a junction dot are separate nets.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /SPI sensors/

File: 06_SPI_Sensors.kicad_sch

Title: Shared SPI bus / position encoder / IMU

Size: A3

Date: 2026-09-15

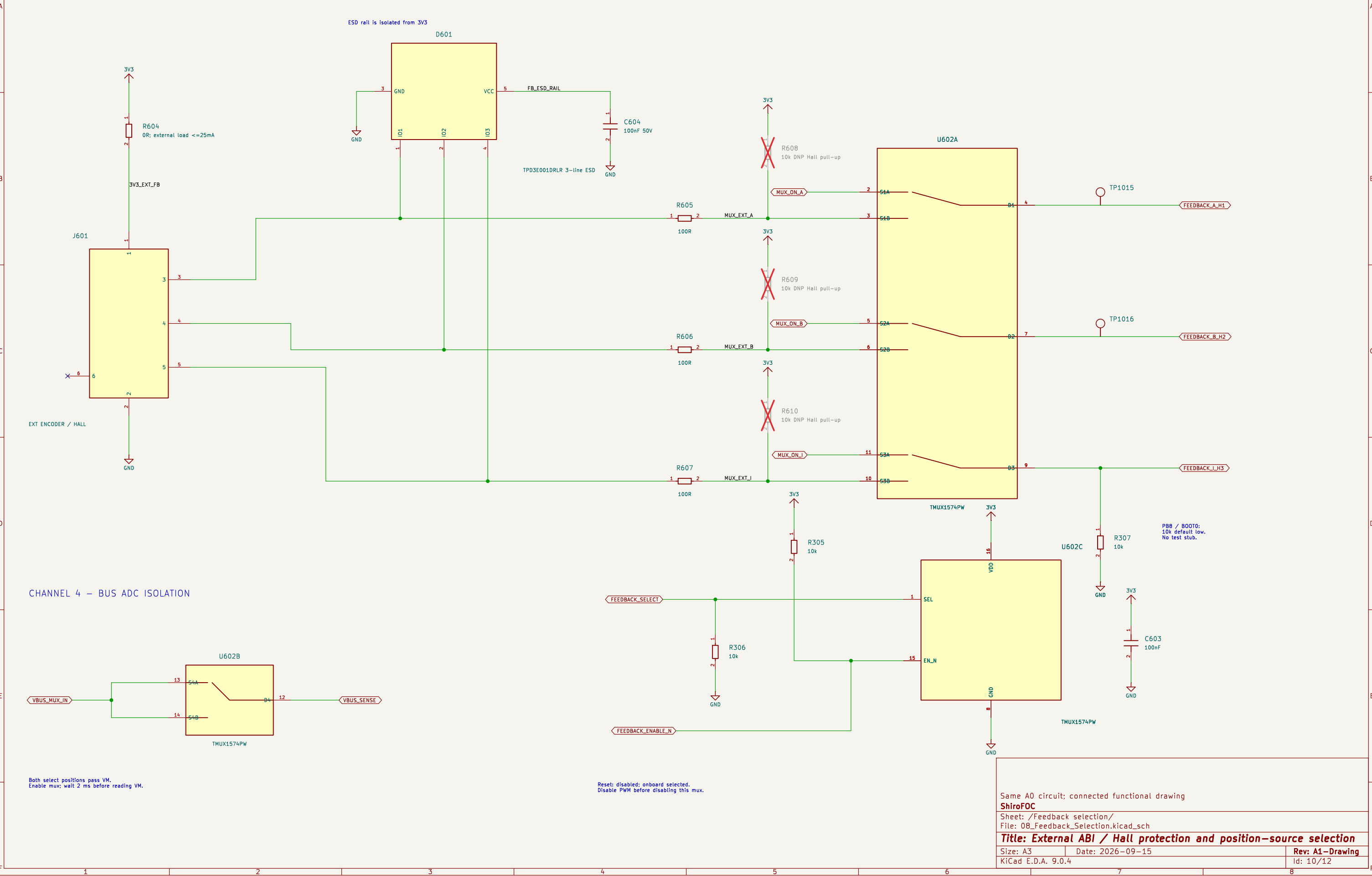
Rev: A1-Drawing

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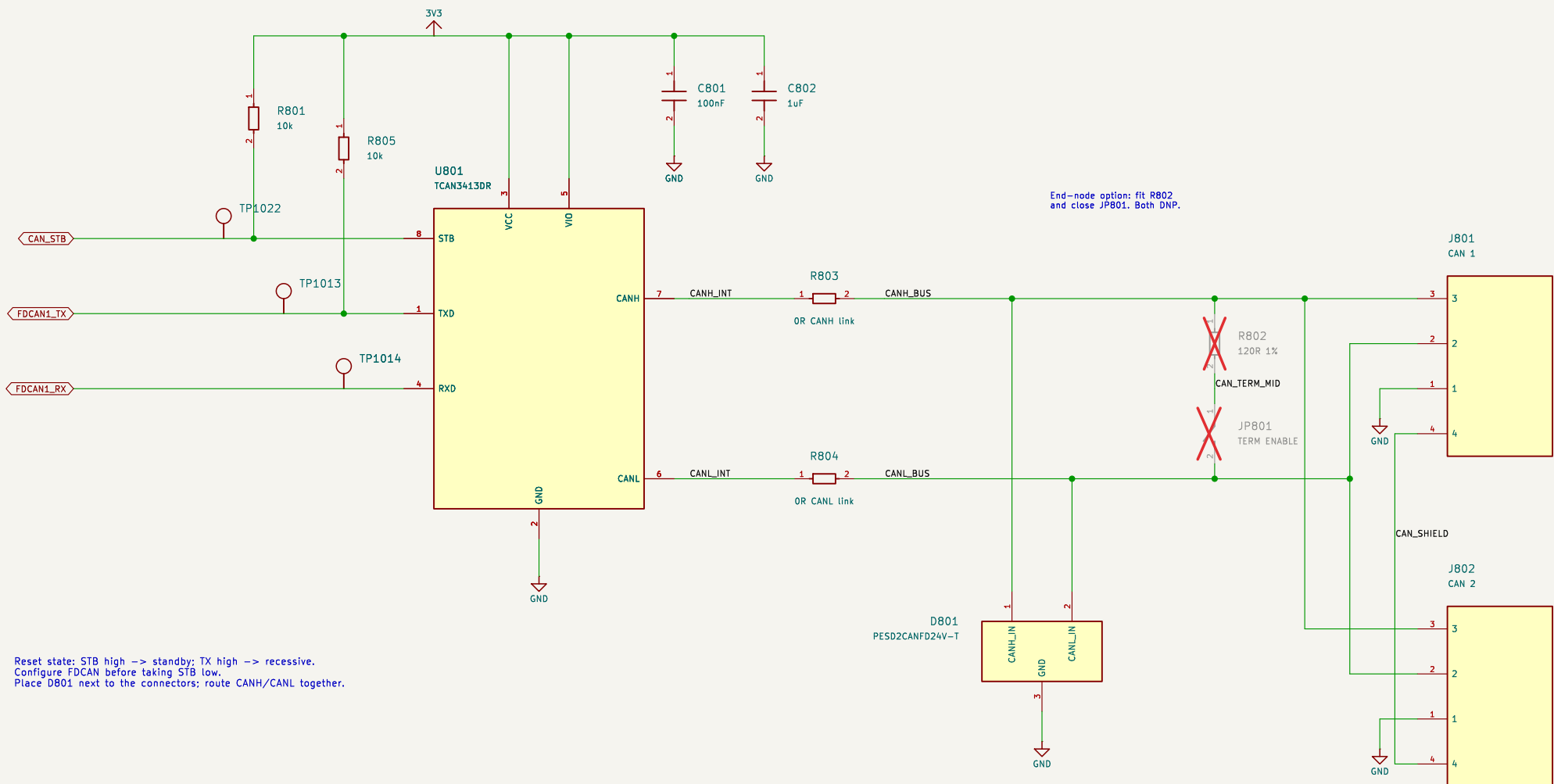
Id: 9/12

ONBOARD OR EXTERNAL POSITION FEEDBACK -> TIM4

External signals must stay within 0–3.3 V, including when board power is off. DNP pull-ups are for open-drain Hall outputs.



CAN: controller -> transceiver -> protected daisy-chain bus



Both headers: 1 GND | 2 CANL | 3 CANH | 4 shield/drain. Pin 4 carries no supply.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /CAN interface/

File: 08_CAN.kicad_sch

Title: CAN / CAN-FD interface

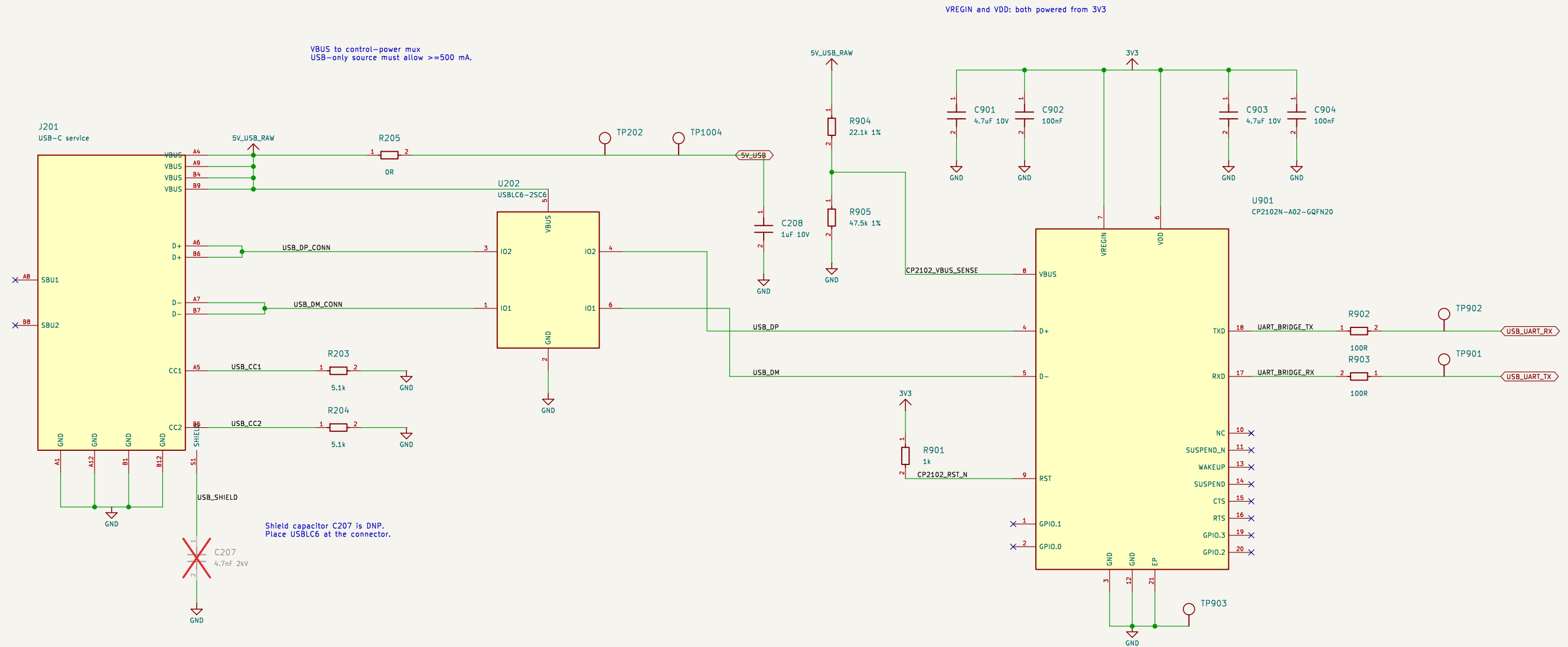
Size: A4	Date: 2026-09-15
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KiCad E.D.A. 9.0.4

Rev: A1—Drawing

Id: 11/12

USB service: connector, protection and bridge in one circuit



LAYOUT: route the USB pair continuously from J201 through U202 to U901; avoid stubs.
C901/C902 belong at VREGIN; C903/C904 belong at VDD. Both pin groups use the same 3V3 rail.

UART labels are named from the MCU point of view.
TXD from U901 drives MCU RX. RTS/CTS are unused.
BOOT0 and reset remain manual controls on the MCU sheet.

Same A0 circuit; connected functional drawing

ShiroFOC

Sheet: /USB service/

File: 09_USB_Service.kicad_sch

Title: USB-C service / UART bridge

Size: A3

Date: 2026-09-15

Rev: A1-Drawing

KiCad E.D.A. 9.0.4

Id: 12/12